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ORIGINAL RESEARCH

Methodological and reporting rigor in noninferiority and equivalence trials for multimodality cancer treatment: lessons from breast cancer radiotherapy

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Abstract

Objectives: To investigate the methodological and reporting quality of noninferiority (NI) or equivalence trials in breast cancer radiotherapy, providing suggestions for future NI/equivalence trials.

Study Design and Setting: Prospective phase III randomized controlled trials comparing different radiation modalities in breast cancer and designed as NI/equivalence were identified. Extracted data included the trial design, analysis, and reporting characteristics. The relationship between trial numbers and publication year was assessed. Trials with prespecified NI margins as absolute risk differences (ARDs) were reevaluated using margins as relative risks.

Results: Twenty-one studies were included. The number of publications increased over time. Trial interventions mainly involved dose fractionation and radiation volume. The primary end point was local or locoregional recurrence in 15, toxicity in 5, and both in 1 study. Reporting gaps included the following: not specifying the trial as NI/equivalence in the title/abstract ($n = 6$); inadequate justification for the NI/equivalence design rationale ($n = 10$) or margins ($n = 12$); absence of both intention-to-treat and per-protocol analysis ($n = 12$); and no reporting of P values for NI/equivalence tests ($n = 12$) or margins with CIs ($n = 5$). Fifteen studies failed to meet their planned accrual target, mostly owing to overestimation of event rates in the control group. Among 8 trials with 9 comparisons claiming NI with prespecified margins as ARDs, 4 comparisons were classified as inconclusive when using the margins as relative risks.

Conclusion: Recently, NI/equivalence trials have dramatically increased in breast cancer radiotherapy; however, there is substantial room for improvement in their methodological and reporting quality. © 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Keywords: Noninferiority; Equivalence; Randomized controlled trial; Breast cancer; Radiotherapy; Methodological quality; Reporting quality

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What is new?

Key findings

- In this systematic review of 21 noninferiority (NI) or equivalence trials in breast cancer radiotherapy, we observed an increasing annual publication rate over time. However, some crucial gaps still remain in their methodological and reporting quality. Accordingly, we have proposed some essential elements for transparent and reproducible reporting to enhance result robustness and reliability.

What this adds to what is known?

- There is substantial room for improvement in the methodological and reporting quality of NI/equivalence trials. Better adherence to existing guidelines, as well as targeted efforts to address the deficiencies identified in this review, will be essential to enhance the robustness and reliability of results.

What is the implication, what should change now?

- Further efforts are also needed to establish clearer, more comprehensive guidelines and implement more stringent publication requirements, ultimately ensuring greater transparency, reproducibility, and impact of NI or equivalence trials in advancing patient care.

1. Introduction

Noninferiority (NI) trials are designed to determine whether an experimental treatment is not significantly worse than the standard treatment by a small, predefined, and clinically acceptable margin. The goal is to demonstrate that the new treatment performs comparably to the standard treatment while offering potential advantages such as lower toxicity, reduced cost, or greater convenience. These designs are particularly valuable when using a placebo or inactive control group is unethical, challenging, or impractical [1−3]. Similarly, equivalence trials share many design principles with NI trials but seek to demonstrate that the new treatment is neither worse nor better than the standard treatment within a predefined margin. Equivalence trials are typically employed when it is essential to establish that the new treatment is identically effective, rather than merely "not worse," than the standard treatment [2,3].

Over the past 2 decades, the number of NI and equivalence trials has grown substantially across various medical disciplines, particularly in oncology and cardiovascular research [4−6]. In breast cancer radiotherapy, advancements in radiotherapy technologies, such as hypofractionation, have driven efforts to optimize treatment delivery. Randomized phase III NI and equivalence trials have provided high-level evidence to inform clinical practice in this context [7−12]. Consequentially, it is crucial for investigators and clinicians to fully understand the design, analysis, and interpretation of NI/equivalence trials.

Compared to superiority trials, NI and equivalence trials present unique challenges, particularly regarding the choice and justification of NI or equivalence margins, which can complicate interpretation. In addition, practical challenges — such as inadvertent enrollment, loss to follow-up, treatment crossover or nonadherence, and the use of concomitant medications — can reduce observed differences between treatments and lead to misleading conclusions of NI or equivalence [4]. Tailored guidelines, such as those developed by the Consolidated Standards of Reporting Trials (CONSORT) group and the US Food and Drug Administration (FDA), aim to improve the quality of study design, reporting and interpretation [1−3]. Nevertheless, systematic reviews have highlighted persistent deficiencies and gaps in the design, reporting, and interpretation of NI and equivalence trials, both in oncology [5,13,14] and other medical fields [4,6,15−17].

In this systematic review, we examined the methodology and reporting quality of published prospective phase III randomized controlled trials (RCTs) designed or interpreted as NI or equivalence between radiotherapy modalities in nonmetastatic breast cancer. Our objectives were to analyze temporal trends, evaluate the quality of statistical design and reporting, and assess the concordance between trial design and interpretation. Based on our findings, we provide recommendations to improve the design, reporting, and interpretation of future NI/equivalence trials.

2. Materials and methods

2.1. Search strategy and selection criteria

We performed a systematic search in the PubMed, EMBASE, and Cochrane library for studies published in the English between January 1, 2001, and December 31, 2022. Search terms included "breast neoplasms," "randomized controlled trial," "noninferiority or equivalence," and "radiotherapy," along with relevant synonyms. The detailed search strategy is provided in Supplemental eTables 1−3. In addition, reference lists of included studies were manually reviewed to identify further eligible studies.

Studies were included if they met the following criteria: (1) patients with nonmetastatic breast cancer treated with

radiotherapy; (2) comparisons between two or more radiotherapy interventions (eg, dose fractionation, whole breast irradiation [WBI] vs. partial breast irradiation [PBI]); (3) and phase III RCTs designed or interpreted as NI or equivalence trials. To preserve methodological comparability, we graded methods using the initial publication that reported the primary end point, and we cataloged major follow-up analyses for debated trials in Supplemental eTable 4 indicating whether any later publications altered the original NI or equivalence inference. Studies were excluded if they were duplicate publications, ongoing studies, conference abstracts, or did not report primary end point results.

2.2. Data extraction

Data extraction was conducted independently by two reviewers (W.S.J. and L.C.N.) using a standardized form (Supplementary Material 2). Discrepancies were adjudicated by two senior authors (S.L.W. and C.H.) and resolved through consensus among all authors when necessary. Extracted data included the following: (1) basic trial characteristics; (2) trial design and analysis details; (3) reporting characteristics for the primary end point; (4) and additional interpretations (detailed in Supplemental eTable 5). Design characteristics were based on the trials' original design considerations. NI/equivalence margins expressed as hazard ratio (HR) or odds ratio were classified as relative risks. When coprimary end points were used, power calculations focused on oncologic efficacy over toxicity. For missing data, publicly accessible trial protocols or design papers were reviewed to supplement information to supplement information.

2.3. Statistical analysis

Categorical variables were summarized as frequencies (percentage), while nonnormally distributed continuous variables were described as median with interquartile range (IQR). Simple linear regression was employed to assess the relationship between the number of published NI/equivalence trials and publication year.

To evaluate whether converting NI margins from absolute risk differences (ARDs) to relative risks would impact trial conclusions, both prespecified and observed margins as relative risks were calculated using Formula (1). Relative risk and corresponding CIs were derived using Formulas (2,3):

$$Margin\ as\ relative\ risk = \frac{event\ rate\ in\ the\ control\ group + pre - specified\ margin\ as\ ARD}{event\ rate\ in\ the\ control\ group} \quad (1)$$

$$relative\ risk = \frac{n1/X1}{n2/X2} \quad (2)$$

$$CI = \frac{n1/X1}{n2/X2} * e^{\pm\ z*sqrt((1/n1+1/n2-1/X1-1/X2))} \quad (3)$$

Here *n1* and *n2* represent the number of primary end point events in the experimental and control group, respectively, while *X1* and *X2* denote the total number of patients in each group. The CI size (z) with determined based on the two-sided α level provided in each study.

All statistical analyses were performed using R (version 3.6.3; http://www.r-project.org/; R Foundation for Statistical Computing, Vienna, Austria).

3. Results

3.1. Trial identification and temporal trends

The database search identified 1617 records, of which 1027 titles or abstracts were screened after removing duplicates. Following screening, 46 full-texts articles were assessed for eligibility, and 16 studies met the inclusion criteria. Five additional studies were included for the following reasons: One study [18] explicitly identified as an NI design in a subsequent long-term publication [19], two studies [20,21] mentioned their NI design only in the full text, and two [22,23] were widely interpreted as NI trials, as confirmed by author consensus. Ultimately, 21 studies were included in the analysis (Fig 1).

As shown in Supplemental eFigure 1, these studies were published between 2002 and 2022, with most (n = 16, 76.2%) appearing after 2012. A significant increase in the publication rate of NI/equivalence trials over time was observed (P = .031 for linear trend).

3.2. Trial characteristics

Table 1 summarizes the basic characteristics and statistical design considerations. Two-thirds of the studies (n = 14) were multicenter and published in high-impact journals such as *Lancet*, *J Clin Oncol*, and *Lancet Oncol*. Most trials (n = 13, 61.9%) were conducted in the UK/Europe. Trial interventions primarily addressed dose fractionation (n = 9, 42.9%), PBI vs. WBI (n = 9, 42.9%), and tumor bed boost (n = 3, 14.3%). Study protocols or design papers were available for 12 (57.1%) studies, and 17

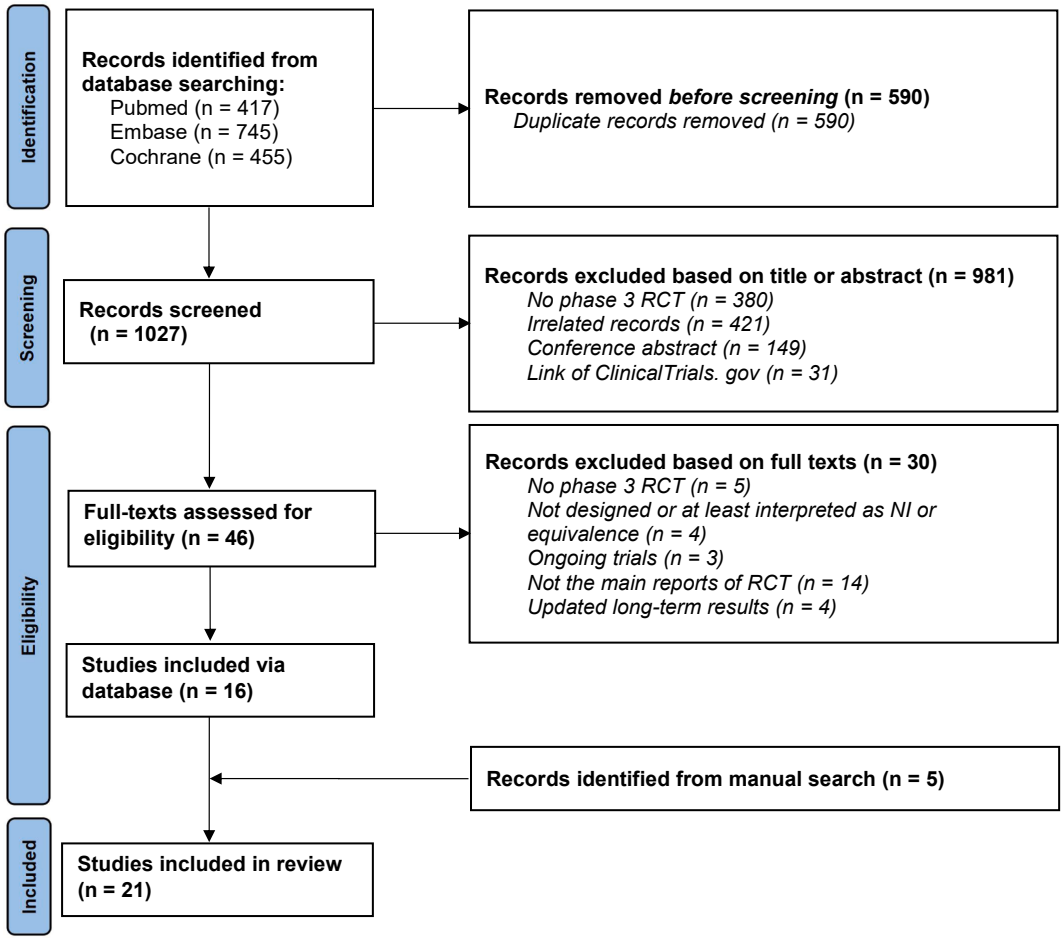

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) diagram summarizing the literature review. RCT, randomized controlled studies; NI, noninferiority.

(81.0%) studies were registered on ClinicalTrials.gov. The majority (95.2%) received public funding. However, six (28.6%) studies did not clearly specify their NI/equivalence design in the title or abstract.

The median recruitment period and follow-up duration were 5.3 years (IQR: 3.4−7.6) and 6.0 years (IQR: 5.1−7.5), respectively. The median number of randomized participants was 1305 (IQR: 511−2175), with a planned sample size of 1000 (IQR: 599−1968). Supplemental eTable 6 provides details on expected and observed follow-up duration, number of events, and sample size.

Eight (38.1%) studies exceeded their planned sample size [10,12,22−27], which two (9.5%) underaccrued by more than 10% [21,28] due to early trial cessation. Three (14.3%) studies were stopped early - two due to new external findings [7,21] and one for an unknown reason [28].

The anticipated benefits of experimental regimens were explicitly stated in 11 (52.4%) studies and inferred from full-texts in 7 (33.3%) studies. Benefits included greater convenience, lower costs, shorter treatment times, and more efficient resource utilization.

3.3. Trial design considerations

Eighteen studies (85.7%) were designed as NI trials, while three (14.3%) were equivalence trials. Most trials (n = 17, 81.0%) had two arms, while four (19.0%) had three arms. Among three-arm studies, pairwise comparisons utilized Bonferroni corrections [11], stricter substudy cutoffs [29], or lacked specified adjustments [7,22]. Randomization schemes included stratified (n = 16, 76.2%), simple (n = 3, 14.3%), and block (n = 2, 9.5%) randomization.

Twenty studies (95.2%) specified α levels and statistical power. One-sided α levels of 0.025 (n = 9, 45.0%), 0.05 (n = 10, 50.0%), and 0.1 (n = 1, 5.0%) were most common. Power levels ranged from ≥79% to ≥90%, with 25% of studies designed with at least 90% power.

Prespecified margins were based on ARDs in 18 (85.7%) studies and relative risks in 2 (9.5%). Among studies using ARDs, 6 (33.3%) provided equivalent relative risks margins. Only 9 (42.9%) studies justified their prespecified margins, and 15 (71.4%) studies provided a justification

Table 1. Trial basic characteristics and design considerations

Characteristics	Trials (n = 21)
Basic characteristics	
Journal	
Lancet	7 (33.3)
J Clin Oncol	4 (19.0)
Lancet Oncol	3 (14.3)
Int J Radiat Oncol Biol Phys	3 (14.3)
Radiother Oncol	1 (4.8)
Breast Cancer Res	1 (4.8)
J Natl Cancer Inst	1 (4.8)
Eur J Cancer	1 (4.8)
Multicenter trial	
Yes	14 (66.7)
No	7 (33.3)
Country	
UK/Europe	13 (61.9)
International	3 (14.3)
China	2 (9.5)
USA	2 (9.5)
Canada	1 (4.8)
Trial intervention	
Dose fractionation	9 (42.9)
PBI vs WBI	9 (42.9)
Tumor bed boost	3 (14.3)
Study protocol or design paper available	
Yes	12 (57.1)
Not found	9 (42.9)
Registration at clinicaltrials.gov	
Yes	17 (81.0)
Not found	4 (19.0)
Source of funding	
Public	20 (95.2)
Public and industry	1 (4.8)
Specifying NI/equivalence trials in title or abstract	
Yes	15 (71.4)
No	6 (28.6)
Recruitment period, median (IQR), years	5.3 (3.4–7.6)
Follow-up time, median (IQR), years	6.0 (5.1–7.5)
Number of patients randomized, median (IQR)	1305 (511–2175)
Planned sample size, median (IQR)	1000 (599–1968)
Stopped early	
Yes	3 (14.3)
No	18 (85.7)
Anticipated benefits	
Clearly indicated	11 (52.4)
Inferred from full-texts	7 (33.3)
Neither stated nor able to be inferred	3 (14.3)
Trial design considerations	
Designed as	
Noninferiority	18 (85.7)
Equivalence	3 (14.3)

(*Continued*)

Table 1. Continued

Characteristics	Trials (n = 21)
Designed as	
Two-arm phase III	17 (81.0)
Three-arm phase III	4 (19.0)
Randomization	
Stratified randomization	16 (76.2)
Simple randomization	3 (14.3)
Block randomization	2 (9.5)
α level (one-sided equivalent)	
0.025	9 (42.9)
0.05	10 (47.6)
0.10	1 (4.8)
Not mentioned	1 (4.8)
Planned power	
$\geq$ 90%	5 (23.8)
80%–89%	14 (66.7)
$<$ 80%	1 (4.8)
Not mentioned	1 (4.8)
Hypothesis based on prespecified margins as	
ARDs	18 (85.7)
Not provided margins as relative risks	12 (66.7)
Provided equivalent margins as hazard ratio converted from ARDs	5 (27.8)
Provided equivalent margins as relative risk converted from ARDs	1 (5.6)
Relative risks (both were hazard ratio)	2 (9.5)
Not provided margins as ARDs	1 (50.0)
Provided equivalent margins as ARDs converted from relative risks	1 (50.0)
Not mentioned	1 (4.8)
Justification for the choice of margin	
Not mentioned	12 (57.1)
Only by consultation with expert	1 (4.8)
Published literature only	2 (9.5)
Author's judgment only	2 (9.5)
Clinicians and patients' agreement	1 (4.8)
Clinicians and patients' agreement; published literature	1 (4.8)
Expert consensus; institutional experiences; published literature; the risk-benefit profile of LR and OS	2 (9.5)
Justification for the choice of event rates in the control group	
Yes	15 (71.4)
No	6 (28.6)

(*Continued*)

Table 1. Continued

Characteristics	Trials (n = 21)
Primary end point	
5-y LR	11 (52.4)
Acute/late toxicities	5 (23.8)
10-y LR	1 (4.8)
5-y LRR	3 (14.3)
2-y LR and cosmetic outcome	1 (4.8)

PBI, partial breast irradiation; WBI, whole breast irradiation; IQR, interquartile range; ARDs, absolute risk differences; LR, local recurrence; LRR, loco-regional recurrence; OS, overall survival; NI, noninferiority.

Data are shown as no. (%).

for the choice of event rates in the control group. Supplemental eTable 7 details the primary end points and margins. Eleven (52.4%) studies chose 5-year local recurrence (LR) as the primary end point: except for one study that did not report the prespecified margin [28], the median prespecified margin as ARD in the remaining ten studies was 2.88% (range: 1.60−6.00). Five (23.8%) studies chose acute/late toxicities as the primary end point: the median prespecified margin as ARD was 10% (range: 5.0−10.0). Other primary end point included 10-year LR (n = 1, 4.8%), 5-year locoregional recurrence (LRR) (n = 3, 14.3%), 2-year LR and cosmetic outcome (n = 1, 4.8%).

3.4. Analysis plan and reporting

Table 2 summarizes the trial analysis and reporting characteristics. Twelve studies (57.1%) used intention-to-treat (ITT) analysis alone, while nine (42.9%) conducted both ITT and per-protocol (PP) analyses, with consistent results. Most conclusions (n = 10, 47.6%) were based on observed ARDs, while others used HRs (n = 3, 14.3%) or both (n = 5, 23.8%).

Sixteen studies (76.2%) reported primary end points with CIs; 15 used two-sided CIs, with most reporting two-sided 95% CIs. Notably, five (23.8%) studies were

Table 2. Trial analysis and reporting characteristics regarding the respective primary end point

Claimed or interpreted as NI or equivalence in the article	Yes (n = 15)	No (n = 2)	Insufficient (n = 4)	Total (n = 21)
Analysis populations				
Only ITT	9 (60.0)	0 (0)	3 (75.0)	12 (57.1)
ITT and PP	6 (40.0)	2 (100.0)	1 (25.0)	9 (42.9)
Conclusions based observed margins as				
ARDs	8 (53.3)	1 (50.0)	1 (25.0)	10 (47.6)
Relative risks(all were hazard ratio)	2 (13.3)	1 (50.0)	0 (0)	3 (14.3)
ARDs and relative risks (all were hazard ratio)	5 (33.3)	0 (0)	0 (0)	5 (23.8)
Not mentioned	0 (0)	0 (0)	3 (75.0)	3 (14.3)
Reported margin with CI				
90% CI (1-sided)	1 (6.7)	0 (0)	0 (0)	1 (4.8)
90% CI (2-sided)	3 (20.0)	1 (50.0)	0 (0)	4 (19.0)
95% CI (2-sided)	10 (66.7)	0 (0)	1 (25.0)	11 (52.4)
not reported	1 (6.7)	1 (50.0)	3 (75.0)	5 (23.8)
CI in agreement with type I error used in sample size calculation				
Yes	10 (66.7)	1 (50.0)	0 (0)	11 (52.4)
No	4 (26.7)	0 (0)	1 (25.0)	5 (23.8)
Not reported	1 (6.7)	1 (50.0)	3 (75.0)	5 (23.8)
Results with P value for noninferiority/equivalence test				
Yes	9 (60.0)	0 (0)	0 (0)	9 (42.9)
Not mentioned	6 (40.0)	2 (100.0)	4 (100.0)	12 (57.1)
Met the protocol's original accrual target				
Yes	6 (40.0)	0 (0)	0 (0)	6 (28.6)
No (overestimated event rates in the control group)	9 (60.0)	2 (100.0)	1 (25.0)	12 (57.1)
No (early stop)	0 (0)	0 (0)	3 (75.0)	3 (14.3)

NI, noninferiority; ARDs, absolute risk differences; ITT, intent-to-treat; PP, per-protocol.

Data are shown as no. (%).

designed with a one-sided α level of 0.05, but reported two-sided 95% CIs. Nine studies (42.9%) reported *P* values for NI/equivalence tests.

Among the six studies achieving their accrual targets, all successfully concluded NI/equivalence. Of the 15 studies failing to meet accrual targets, 12 (80.0%) overestimated control group event rates. Nine of these 12 studies ultimately concluded NI/equivalence after protocol amendments or post hoc justifications [9,12,18,20,22–25,27,29–31]. Supplemental eTable 8 provides details on countermeasures.

3.5. Trial interpretations

Fifteen studies (71.4%) claimed NI/equivalence [8–12,18,22–24,26,27,29,31–33], while two (9.5%) did not [25,30], and four (19.0%) were inconclusive due to insufficient sample sizes. For trials did not claim NI/equivalence or were inconclusive, the RTOG 0413 trial narrowly missed the prespecified margin [30].

Among 18 comparisons reassessed using CONSORT statistical criteria, 16 were concordant with the original publications (Supplemental eFig 2). Two were reclassified. In TARGeted Intraoperative radioTherapy Alone (TARGIT-A) trial [27], for 5 year LR, the estimate and CI lay entirely to the right of the null yet within the prespecified absolute NI margin of 2.5 percentage points. In DBCG-PBI [26], for grade 2 to 3 breast induration at 3 years, the estimate and CI lay entirely to the left of the null, indicating superiority of the experimental treatment. Key details and citations for TARGIT-A are summarized in Supplemental eTable 4.

3.6. Overestimated event rates

We further reevaluated the eight studies (nine comparisons) that claimed NI based solely on ARD-based margins (Fig 2A). Six (75%) of these eight studies provided a justification for their choice of event rates in the control group. Seven (77.8%) of these nine comparisons overestimated event rates in the control group. Such overestimation often led to larger relative risks margins, potentially biasing NI conclusions. As shown in Figure 2B and Supplemental eTable 9, when reassessed using relative risk-based margins, four comparisons (44.4%) were reclassified as inconclusive [22,24,27].

4. Discussion

In this systematic review of 21 NI/equivalence trials in breast cancer radiotherapy, we observed an increasing annual publication rate over time. While most studies were multicenter, large-scale, and published in high-impact medical journals, notable discrepancies in recruitment were evident, with eight (38.1%) studies overaccruing and two

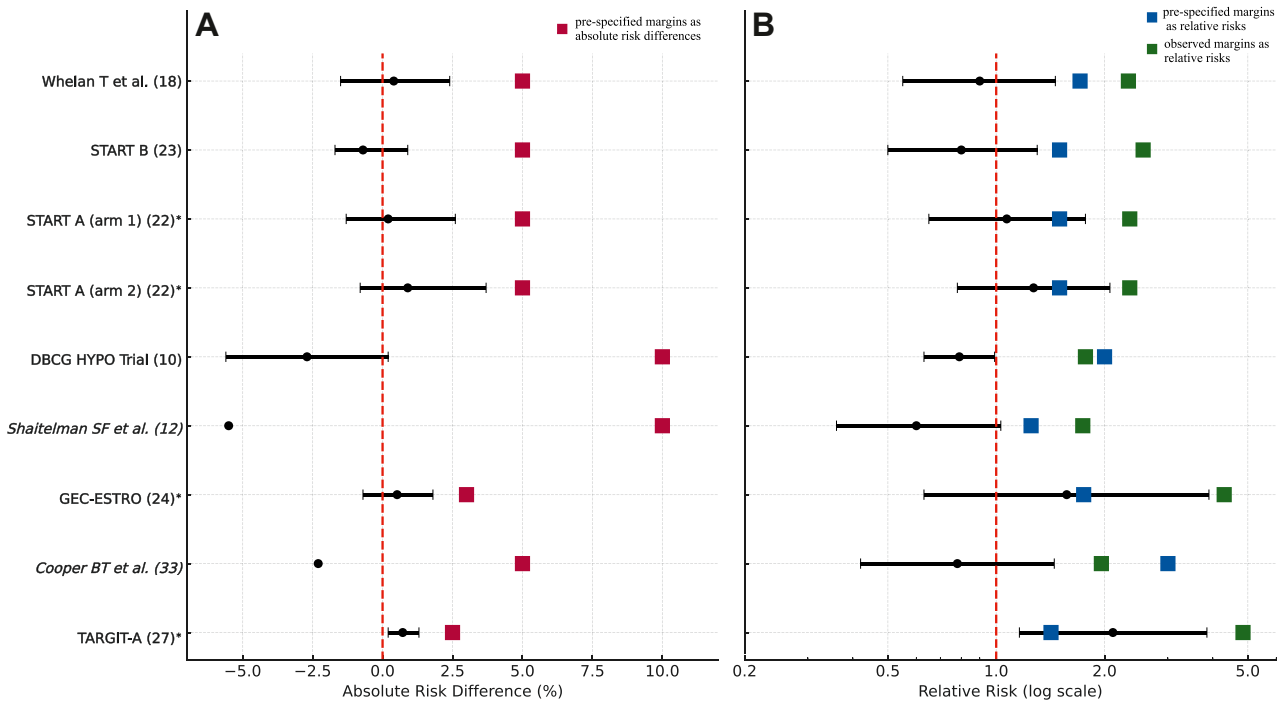

Figure 2. Forest plot of CI for absolute risk differences (A) and relative risks (B) for the nine included comparisons only based on margins as absolute risk differences. The red square represents the prespecified margins as absolute risk differences. The green and blue square represent the observed and prespecified margins as relative risks, respectively. Trial names in *italics* indicate studies that did not provide a complete CI for absolute risk differences. *represents the comparisons that were reclassified as inconclusive. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 3. Key elements for transparent reporting of NI or equivalence trials

Section/topic	Items
Title or abstract	Clear identification as an NI/equivalence trial in the title/abstract.
Background and objectives	Clear statements of the rational for an NI/equivalence design.
Design-hypothesis	Specifying the NI/equivalence hypothesis based on margins as absolute risk differences or relative risks.
Design-margin	Selecting the magnitude of the margin, ideally reflecting statistical and clinical considerations, and patient preferences; clear justifications for the choice of the margins.
Design-sample size	Specifying whether the sample size was calculated using an NI/equivalence criterion and reporting the complete parameters about the sample size calculation (eg, alternative hypothesis, α level, planned power, NI/equivalence, etc.).
Analysis	Performing the analysis in intention-to-treat and per-protocol populations to facilitate reliable conclusions.
Analysis	Testing NI/equivalence based on both absolute and relative risk differences.
Reporting	Reporting the observed margins with CI and *P* values for an NI/equivalence test.
Reporting	Reporting CI in agreement with the type I error used in the sample size calculation.
Interpretations	Aligning the conclusions with the study findings according to statistical aspects of the CONSORT criteria.
Interpretations	Clearly stating the overestimation of event rates in the control group when interpreting the results, especially for trials failing to demonstrate statistical significance.
Others	The study protocol or design paper is available online.

NI, noninferiority; CONSORT, Consolidated Standards of Reporting Trials.

(9.5%) studies underaccruing by more than 10% of the planned sample sizes. The trial interventions predominantly focused on dose fractionation and PBI *vs* WBI. Most studies selected 5-year LR as the primary end point and provided sufficient details to replicate sample size calculation. Despite this, key reporting gaps persisted, including the following: (i) failure to specify the trial as an NI/equivalence study in the title or abstract; (ii) unavailability of the study protocol or design paper; (iii) insufficient justifications for the NI/equivalence design rationale and margins, particularly when wide margins were used; (iv) lack of both ITT and PP analyses; and (v) inadequate reporting of statistical inferences, such as *P* values for NI/equivalence tests or CI consistent with the statistical design. A frequent issue across trials was the overestimation of the control group event rates, which can unintentionally affect the interpretation of NI/equivalence outcomes, especially when margins are based on ARDs or relative risks. Addressing such overestimation requires prospective countermeasures, including blinded interim analyses or event-driven follow-up durations, to ensure robust statistical conclusions [34].

Building upon the extension of CONSORT 2010 checklist [3] for NI/equivalence trials, we developed a supplementary list of key elements to enhance transparency and reproducibility in reporting NI or equivalence randomized trials (Table 3). While many items reiterate existing CONSORT guidance, our findings suggest that strict adherence to these elements would have greatly improved the interpretability of NI/equivalence trials in breast cancer radiotherapy over the past decade.

Five additional studies identified through manual searching underscore the importance of transparent reporting [18,20–23]. These studies were not retrieved via database searches because they did not identify as NI/equivalence trials in their titles or abstracts, violating a key CONSORT recommendation [3]. While two clarified their NI/equivalence design within the methods section [20,21], others failed to comply fully with CONSORT 2010 guidance, leading to interpretive variability and highlighting the critical need for uniform and transparent reporting standards. Notably, most of these trials were conducted at a time when guidance about reporting NI or equivalence trials was not yet widely disseminated.

The growing prevalence of NI/equivalence trials in breast cancer radiotherapy aligns with advancements in radiotherapy technology and the broader trend of evaluating trade-offs between efficacy and nonefficacy benefits, such as reduced treatment costs, shorter durations, and increased convenience. However, our review revealed that 14.3% of studies (all published after 2016) lacked clear justifications for NI/equivalence designs, a figure consistent with earlier reviews across all cancer types [35] but significantly better than the 39% observed in drug trials [36]. Clear justification of design rationale in primary publications remains essential to enhance understanding of benefit-risk trade-offs.

The choice of NI margins, whether ARDs or relative risks, remains a contentious issue [37]. While ARDs offer intuitive interpretations of risk changes relevant to patients [38], they rely heavily on accurate assumptions about control group event rates. Overestimations can lead to permissive relative risk margins and, in turn, affect NI claims [39]. In our review, four of nine comparisons claiming NI using ARD margins were reclassified as inconclusive when reassessed using relative risk margins. This highlights the need for balanced consideration of ARDs and relative risks,

particularly in contexts like breast cancer radiotherapy, where low LRR rates complicate statistical power calculations. A clinically justified ARD margin, as seen in the TARGIT-A trial [27], can sometimes offer a reasonable compromise but requires careful selection to avoid overly arbitrary or wide margins. Efforts to establish consensus on margin selection are ongoing. Approaches such as the Delphi method [40], which integrates expert opinion, and patient-inclusive methodologies [41] offer promising strategies for identifying clinically acceptable margins. Further exploration and standardization are needed to align statistical and clinical perspectives.

Accurate estimation of control group event rates remains crucial. Over half of the trials (57.1%) in our review overestimated these rates, consistent with findings in other fields [39]. Given the long durations and evolving efficacy of treatments, overestimation introduces challenges in trial design and interpretation. Blinded interim analyses or event-driven designs have been suggested as potential solutions [17], yet practical guidelines addressing this issue are lacking.

For end point analysis, both ITT and PP analyses are essential to ensure robust conclusions, as recommended by CONSORT and FDA guidelines [1−3]. In our review, less than half of the studies performed both analyses, with most relying solely on PP analyses. While ITT analyses may increase type I errors in NI trials, PP analyses risk postrandomization biases [4]. Adopting both approaches is critical to mitigating these limitations and improving result reliability.

Most comparisons in our review were consistent with the original publications when reassessed using CONSORT statistical criteria [2,3]. Two were reclassified, highlighting distinct methodological issues. In TARGIT-A [27], for 5 year LR, the CI lay entirely within the prespecified absolute NI margin yet entirely to the right of the null, indicating statistical inferiority for LR despite meeting the NI criterion. This paradox is more likely when absolute margins are used and the observed control risk is lower than anticipated, which makes the absolute margin relatively permissive. In DBCG-PBI [26], the CI lay entirely to the left of the null, indicating superiority of the experimental treatment. We cataloged the long term follow-up publications about TARGIT-A trial in Supplemental eTable 4, and noted whether later publications alter the original NI claim. Together, these examples reinforce the need to prespecify an interpretable margin and scale, anticipate departures between assumed and observed event rates.

To our knowledge, this is the first systematic review assessing the methodological and reporting quality of NI/equivalence trials in breast cancer radiotherapy. While we identified key gaps and proposed elements to enhance reporting, several limitations should be noted. First, our database search may have missed studies due to incomplete reporting of NI/equivalence design in titles or abstracts, a gap we aim to highlight. Second, defining anticipated benefits relied on reviewers' clinical judgment, introducing potential subjectivity. Third, our focus on breast cancer radiotherapy limits the generalizability of findings to other fields.

5. Conclusion

NI and equivalence trials in breast cancer radiotherapy have dramatically increased in recent years; however, substantial room remains for improvement in their methodology and reporting quality. Better adherence to existing guidelines, as well as a focused effort to address the deficiencies identified in this review, will be essential to enhance the robustness and reliability of results. Further efforts are also needed to establish clearer, more comprehensive guidelines and implement more stringent publication requirements, ultimately ensuring greater transparency, reproducibility, and impact of NI/equivalence trials in advancing patient care.

CRediT authorship contribution statement

Shi-Jia Wang: Writing − original draft, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Chun-Nan Liu:** Validation, Methodology, Investigation, Data curation. **Yu Tang:** Validation, Supervision, Investigation. **Hao Jing:** Validation, Supervision, Investigation. **Hui Fang:** Validation, Supervision, Investigation. **Yi-Rui Zhai:** Validation, Supervision, Investigation. **Si-Ye Chen:** Validation, Supervision, Investigation. **Guang-Yi Sun:** Validation, Supervision, Investigation. **Xu-Ran Zhao:** Validation, Supervision, Investigation. **Yu-Chun Song:** Validation, Supervision, Investigation. **Yong-Wen Song:** Validation, Supervision. **Yue-Ping Liu:** Validation, Supervision. **Bo Chen:** Validation, Supervision. **Shu-Nan Qi:** Validation, Supervision. **Yuan Tang:** Validation, Supervision. **Ning-Ning Lu:** Validation, Supervision. **Wen-Wen Zhang:** Validation, Supervision. **Julia R. White:** Writing − review & editing, Validation, Supervision. **Ye-Xiong Li:** Writing − review & editing, Validation, Supervision, Investigation. **Shu-Lian Wang:** Writing − review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Chen Hu:** Writing − review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

There are no competing interests for any author.

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Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jclinepi.2025.112012.

Data availability

All data are incorporated into the article and its online Supplementary Material and will be available with publication.

References

[1] U.S. Department of Health and Human Services, Food and Drug Administration. Non-inferiority clinical trials to establish effectiveness—guidance for industry. 2016. Available at: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/non-inferiority-clinical-trials. Accessed March 1, 2023.

[2] Piaggio G, Elbourne DR, Altman DG, Pocock SJ, Evans SJW, CONSORT Group. Reporting of noninferiority and equivalence randomized trials: an extension of the CONSORT statement. JAMA 2006;295:1152−60.

[3] Piaggio G, Elbourne DR, Pocock SJ, Evans SJW, Altman DG, CONSORT Group. Reporting of noninferiority and equivalence randomized trials: extension of the CONSORT 2010 statement. JAMA 2012;308:2594−604.

[4] Mauri L, D'Agostino RB. Challenges in the design and interpretation of noninferiority trials. In: Drazen JM, Harrington DP, McMurray JJV, et al, editors. N Engl J Med 2017;377:1357−67.

[5] Tanaka S, Kinjo Y, Kataoka Y, Yoshimura K, Teramukai S. Statistical issues and recommendations for noninferiority trials in oncology: a systematic review. Clin Cancer Res 2012;18:1837−47.

[6] Bikdeli B, Welsh JW, Akram Y, Punnanithinont N, Lee I, Desai NR, et al. Noninferiority designed cardiovascular trials in highest-impact journals. Circulation 2019;140:379−89.

[7] Yarnold J, Ashton A, Bliss J, Homewood J, Harper C, Hanson J, et al. Fractionation sensitivity and dose response of late adverse effects in the breast after radiotherapy for early breast cancer: long-term results of a randomised trial. Radiother Oncol 2005;75:9−17.

[8] Wang S-L, Fang H, Song Y-W, Wang WH, Hu C, Liu YP, et al. Hypofractionated versus conventional fractionated postmastectomy radiotherapy for patients with high-risk breast cancer: a randomised, non-inferiority, open-label, phase 3 trial. Lancet Oncol 2019;20:352−60.

[9] Wang S-L, Fang H, Hu C, Song YW, Wang WH, Jin J, et al. Hypofractionated versus conventional fractionated radiotherapy after breast-conserving surgery in the modern treatment era: a multicenter, randomized controlled trial from China. J Clin Oncol 2020;38:3604−14.

[10] Offersen BV, Alsner J, Nielsen HM, Jakobsen EH, Nielsen MH, Krause M, et al. Hypofractionated versus standard fractionated radiotherapy in patients with early breast cancer or ductal carcinoma in situ in a randomized phase III trial: the DBCG HYPO trial. J Clin Oncol 2020;38:3615−25.

[11] Murray Brunt A, Haviland JS, Wheatley DA, Sydenham MA, Alhasso A, Bloomfield DJ, et al. Hypofractionated breast radiotherapy for 1 week versus 3 weeks (FAST-Forward): 5-year efficacy and late normal tissue effects results from a multicentre, non-inferiority, randomised, phase 3 trial. Lancet 2020;395:1613−26.

[12] Shaitelman SF, Lei X, Thompson A, Schlembach P, Bloom ES, Arzu IY, et al. Three-year outcomes with hypofractionated versus conventionally fractionated whole-breast irradiation: results of a randomized, noninferiority clinical trial. J Clin Oncol 2018;36:JCO1800317.

[13] Paesmans M, Grigoriu B, Ocak S, Roelandts M, Lafitte JJ, Holbrechts S, et al. Systematic qualitative review of randomised trials conducted in nonsmall cell lung cancer with a noninferiority or equivalence design. Eur Respir J 2015;45:511−24.

[14] Wayant C, Ross A, Vassar M. Methodological quality of oncology noninferiority clinical trials. Crit Rev Oncol Hematol 2020;149:102938.

[15] Aberegg SK, Hersh AM, Samore MH. Empirical consequences of current recommendations for the design and interpretation of noninferiority trials. J Gen Intern Med 2018;33:88−96.

[16] Rehal S, Morris TP, Fielding K, Carpenter JR, Phillips PPJ. Non-inferiority trials: are they inferior? A systematic review of reporting in major medical journals. BMJ Open 2016;6:e012594.

[17] Macaya F, Ryan N, Salinas P, Pocock SJ. Challenges in the design and interpretation of noninferiority trials. J Am Coll Cardiol 2017;70:894−903.

[18] Whelan T, MacKenzie R, Julian J, Levine M, Shelley W, Grimard L, et al. Randomized trial of breast irradiation schedules after lumpectomy for women with lymph node-negative breast cancer. J Natl Cancer Inst 2002;94:1143−50.

[19] Whelan TJ, Julian JA, Shelley W, et al. Long-term results of hypofractionated radiation therapy for breast cancer. New Engl J Med 2010;362:513−20.

[20] Livi L, Meattini I, Marrazzo L, Simontacchi G, Pallotta S, Saieva C, et al. Accelerated partial breast irradiation using intensity-modulated radiotherapy versus whole breast irradiation: 5-year survival analysis of a phase 3 randomised controlled trial. Eur J Cancer 2015;51:451−63.

[21] Polgár C, Fodor J, Major T, Németh G, Lövey K, Orosz Z, et al. Breast-conserving treatment with partial or whole breast irradiation for low-risk invasive breast carcinoma–5-year results of a randomized trial. Int J Radiat Oncol Biol Phys 2007;69:694−702.

[22] START Trialists' Group, Bentzen SM, Agrawal RK, Aird EGA, Barrett JM, Barrett-Lee PJ. The UK standardisation of breast radiotherapy (START) trial A of radiotherapy hypofractionation for treatment of early breast cancer: a randomised trial. Lancet Oncol 2008;9:331−41.

[23] START Trialists' Group, Bentzen SM, Agrawal RK, Aird EGA, Barrett JM, Barrett-Lee PJ. The UK standardisation of breast radiotherapy (START) trial B of radiotherapy hypofractionation for treatment of early breast cancer: a randomised trial. Lancet 2008;371:1098−107.

[24] Strnad V, Ott OJ, Hildebrandt G, Kauer-Dorner D, Knauerhase H, Major T, et al. 5-year results of accelerated partial breast irradiation using sole interstitial multicatheter brachytherapy versus whole-breast irradiation with boost after breast-conserving surgery for low-risk invasive and in-situ carcinoma of the female breast: a randomised, phase 3, non-inferiority trial. Lancet 2016;387:229−38.

[25] Veronesi U, Orecchia R, Maisonneuve P, Viale G, Rotmensz N, Sangalli C, et al. Intraoperative radiotherapy versus external radiotherapy for early breast cancer (ELIOT): a randomised controlled equivalence trial. Lancet Oncol 2013;14:1269−77.

[26] Offersen BV, Alsner J, Nielsen HM, Jakobsen EH, Nielsen MH, Stenbygaard L, et al. Partial breast irradiation versus whole breast irradiation for early breast cancer patients in a randomized phase III trial: the Danish breast cancer group partial breast irradiation trial. JCO 2022;40:4189−97.

[27] Vaidya JS, Wenz F, Bulsara M, Tobias JS, Joseph DJ, Keshtgar M, et al. Risk-adapted targeted intraoperative radiotherapy versus whole-breast radiotherapy for breast cancer: 5-year results for local control and overall survival from the TARGIT-A randomised trial. Lancet 2014;383:603−13.

[28] Ciabattoni A, Gregucci F, Fastner G, Cavuto S, Spera A, Drago S, et al. IOERT versus external beam electrons for boost radiotherapy

in stage I/II breast cancer: 10-year results of a phase III randomized study. Breast Cancer Res 2021;23:46.

[29] Coles CE, Griffin CL, Kirby AM, Titley J, Agrawal RK, Alhasso A, et al. Partial-breast radiotherapy after breast conservation surgery for patients with early breast cancer (UK IMPORT LOW trial): 5-year results from a multicentre, randomised, controlled, phase 3, non-inferiority trial. Lancet 2017;390:1048−60.

[30] Vicini FA, Cecchini RS, White JR, Arthur DW, Julian TB, Rabinovitch RA, et al. Long-term primary results of accelerated partial breast irradiation after breast-conserving surgery for early-stage breast cancer: a randomised, phase 3, equivalence trial. Lancet 2019;394:2155−64.

[31] Whelan TJ, Julian JA, Berrang TS, Kim DH, Germain I, Nichol AM, et al. External beam accelerated partial breast irradiation versus whole breast irradiation after breast conserving surgery in women with ductal carcinoma in situ and node-negative breast cancer (RAPID): a randomised controlled trial. Lancet 2019;394:2165−72.

[32] Hörner-Rieber J, Forster T, Hommertgen A, Haefner MF, Arians N, König L, et al. Intensity Modulated Radiation Therapy (IMRT) with simultaneously integrated boost shortens treatment time and is noninferior to conventional radiation therapy followed by sequential boost in adjuvant breast cancer treatment: results of a large randomized phase III trial (IMRT-MC2 trial). Int J Radiat Oncol Biol Phys 2021;109:1311−24.

[33] Cooper BT, Formenti-Ujlaki GF, Li X, Shin SM, Fenton-Kerimian M, Guth A, et al. Prospective randomized trial of prone accelerated intensity modulated breast radiation therapy with a daily versus weekly boost to the tumor bed. Int J Radiat Oncol Biol Phys 2016;95:571−8.

[34] Korn EL, Freidlin B. Interim monitoring for non-inferiority trials: minimizing patient exposure to inferior therapies. Ann Oncol 2018; 29:573−7.

[35] Riechelmann RP, Alex A, Cruz L, Bariani GM, Hoff PM. Non-inferiority cancer clinical trials: scope and purposes underlying their design. Ann Oncol 2013;24:1942−7.

[36] Gyawali B, Tessema FA, Jung EH, Kesselheim AS. Assessing the justification, funding, success, and survival outcomes of randomized noninferiority trials of cancer drugs: a systematic review and pooled analysis. JAMA Netw Open 2019;2:e199570.

[37] Abulizi X, Flandre P. Choice of treatment-effect measures when non-inferiority margins originally defined in absolute difference translated into relative difference influenced the results of clinical trials. J Clin Epidemiol 2018;96:63−72.

[38] Cuzick J, Sasieni P. Interpreting the results of noninferiority trials—a review. Br J Cancer 2022;127:1755−9.

[39] Simonato M, Ben-Yehuda O, Vincent F, Zhang Z, Redfors B. Consequences of inaccurate assumptions in coronary stent noninferiority trials: a systematic review and meta-analysis. JAMA Cardiol 2022; 7:320−7.

[40] Acuna SA, Chesney TR, Amarasekera ST, Baxter NN. Defining non-inferiority margins for quality of surgical resection for rectal cancer: a Delphi consensus study. Ann Surg Oncol 2018;25:3171−8.

[41] Acuna SA, Chesney TR, Baxter NN. Incorporating patient preferences in noninferiority trials. JAMA 2019;322:305.